DAKSH 2026
Build Things That Matter

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TATA CONSULTANCY SERVICES

TCS FOUNDATION
presents

DAKSH AI HACKATHON

Powered by amazon

Category 1: Healthcare & Medical AI

Category 2: EDUCATION & LEARNING

Category 3: AGRICULTURE & SUSTAINABILITY

SASTRA's Annual Tech-Tonic!

Category 4: BUSINESS & E-COMMERCE

Category 5: SMART HOMES

Category 6: CYBERPHYSICAL SYSTEMS

Category 7: ACCESSIBILITY

March 13 - 15, 2026

AI Hackathon Competition Rules & Guidelines

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AI Hackathon
Competition Rules & Guidelines

Competition Structure

Stage 1: Prelims Round - Business Case Evaluation

- Objective: Filter top 15 teams based on business viability and cost-effectiveness
- Submission Type: Business proposal + feasibility analysis
- Duration: 1 week from problem statement release
- Focus: Prove your solution won't bleed money!

Stage 2: Finals Round

- Objective: Develop working solutions with validated economics in a AWS environment provided by the organizers.
- Submission Type: Working demo + cost analysis + documentation
- Duration: 24 hrs for development

Participation Rules

Team Formation

1. Team Size: 2-4 members per team
2. Registration: Each team must register with a unique team name
3. Problem Selection: Each team selects ONE problem statement from the provided list. (One problem within a category, if a category has multiple problems) The teams can aim to solve one or more challenge stream tracks within a problem statement.
4. Team Changes: No team member changes allowed after registration deadline

- Open to all college students (undergraduate and postgraduate)
- Cross-department teams are encouraged
- Faculty mentorship is allowed but faculty cannot be team members
- Each student can be part of only ONE team

Prelims Round Evaluation (100 Points)

Submission Requirements

Teams must submit a Business Proposal Document (Max 6 pages) containing:

1. Problem & Market Analysis

- Problem statement understanding
- Target user/customer segment
- Market size and opportunity
- Current solutions and their limitations

2. Proposed Solution Overview

- High-level solution description
- Key features and functionality
- Technical approach (AI/ML techniques)
- Unique value proposition

3. Cost-Benefit Analysis CRITICAL

Cost per Consumer/Transaction

- Infrastructure costs (compute, storage, bandwidth)
- API costs (if using third-party services)
- Model training and inference costs
- Maintenance and support costs
- Break down to per-user or per-transaction level

Revenue/Value Model

- How will this generate value?
- Pricing strategy (if applicable)
- Cost savings for end users
- ROI timeline

Cost Comparison

- Current solution costs vs. your solution
- Competitive pricing analysis
- Cost optimization strategies

4. Scalability Analysis

Technical Scalability

- Can it handle 100 users? 10,000? 1 million?
- Infrastructure scaling strategy
- Performance degradation analysis
- Bottleneck identification

Cost Scalability

- How does cost per user change with scale?
- Economies of scale opportunities
- Break-even point analysis

5. Implementation Feasibility

Ease of Deployment

- Setup complexity (1-10 scale)
- Infrastructure requirements
- Integration with existing systems
- Time to deployment

Ease of Use

- User experience considerations
- Learning curve for end users
- Maintenance requirements
- Support needs

6. Risk & Mitigation

- Technical risks
- Cost overrun risks
- Adoption barriers
- Mitigation strategies

Prelims Evaluation Criteria

Criteria	Points	Description
Cost Economics	30	Cost per consumer/transaction, pricing model, profitability path, cost optimization strategy
Scalability	20	Technical and cost scalability, performance at scale, infrastructure efficiency
Business Viability	20	Market opportunity, revenue model, competitive advantage, ROI potential
Implementation Feasibility	10	Ease of deployment, integration complexity, maintenance burden, technical capability
Problem Understanding	10	Depth of analysis, user needs identification, market research quality
Ethical and privacy concerns and guardrails	10	Maintaining user data privacy, guardrails to check bot attacks, Jailbreaking etc

Mandatory Requirements for Qualification

Teams MUST demonstrate:

1. Cost per consumer/transaction < competitive alternatives OR clear path to cost reduction
2. Scalability plan showing cost efficiency improves or stabilizes with scale
3. Implementation complexity rated ≤ 7/10
4. No dependency on expensive proprietary tools without justification

Failure to meet these = Automatic disqualification

Finals Round Evaluation (100 Points)

Submission Requirements

1. Working Demo/Prototype

- Functional implementation
- Live demonstration capability
- Performance metrics dashboard

2. Validated Cost Analysis

- Actual costs incurred during development
- Projected costs at different scales (100, 1K, 10K, 100K users)
- Cost breakdown spreadsheet
- Comparison with prelims estimates

3. Source Code & Documentation

- Complete codebase on GitHub/GitLab
- Setup and deployment guide
- Architecture documentation
- API documentation (if applicable)

4. Business Metrics Report

- Performance benchmarks
- Resource utilization metrics
- Cost per transaction/user (actual)
- Scalability test results

5. Final Presentation (15-minute demo + 10-minute Q&A)

- Live demonstration (5 min)
- Cost validation and economics (5 min)
- Scalability proof (3 min)
- Business impact (2 min)

Finals Evaluation Criteria

Criteria	Points	Description
Cost Validation	25	Actual costs match estimates, cost per user is competitive, optimization implemented
Working Demo	20	Functionality, stability, performance, user experience
Scalability Proof	20	Load testing results, cost efficiency at scale, infrastructure optimization
Technical Implementation	15	Code quality, architecture, AI/ML effectiveness
Business Impact	10	Real-world applicability, user value, market potential
Ethical and privacy concerns, guardrails	10	Maintaining user data privacy, guardrails to check bot attacks, Jailbreaking etc

DO's

Business & Economics

- ☑ Calculate EXACT cost per user/transaction
- ☑ Use free-tier services where possible
- ☑ Optimize model size and inference costs
- ☑ • Consider open-source alternatives to paid tools
- ☑ Plan for cost reduction at scale
- ☑ Research competitive pricing
- ☑ Document all cost assumptions

Technical

- ☑ Use efficient algorithms and models
- ☑ Implement caching strategies
- ☑ Optimize database queries
- ☑ Implement auto-scaling
- ☑ Monitor resource usage
- ☑ Use lightweight models where possible
- ☑ Batch processing for efficiency

Implementation

- ☑ Create simple deployment scripts
- ☑ Use containerization (Docker)
- ☑ Provide clear setup instructions
- ☑ Design for easy maintenance
- ☑ Build modular architecture
- ☑ Plan for incremental rollout

DON'Ts

Cost & Economics

- ☒ Don't ignore infrastructure costs
- ☒ Don't rely on expensive proprietary APIs without justification
- ☒ Don't use oversized models when smaller ones suffice
- ☒ Don't ignore data transfer costs
- ☒ Don't assume unlimited free resources
- ☒ Don't create solutions that cost more than the problem
- ☒ Don't ignore operational costs (monitoring, logging, support)
- ☒ Don't use paid services without exploring free alternatives

Scalability

- ☒ Don't design for only 10 users
- ☒ Don't ignore performance bottlenecks
- ☒ Don't assume linear scaling
- ☒ Don't hardcode resource limits
- ☒ Don't skip load testing

Implementation

- ☒ Don't create overly complex architectures
- ☒ Don't require manual intervention for scaling
- ☒ Don't build monolithic systems
- ☒ Don't ignore deployment automation
- ☒ Don't make setup require expert knowledge